Introduction to Data Analysis with R

Day 1: Welcome to the workshop

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Who am I?

💻 **Scientific programmer** @theoretical ecology group
🎓 PhD in **dryland ecology**
👩‍🏫 **Teaching** R, Git, good scientific practice, …

I'm using R a lot for ...

... data analysis

... ecological modelling

... writing documents/websites/presentations

.... workflow automation

...

Who are you?

Optimising **CAR-T cell treatment** using **mathematical modelling**

Improving **microdialysis calibration** approaches.

Survival of **memory B cells**

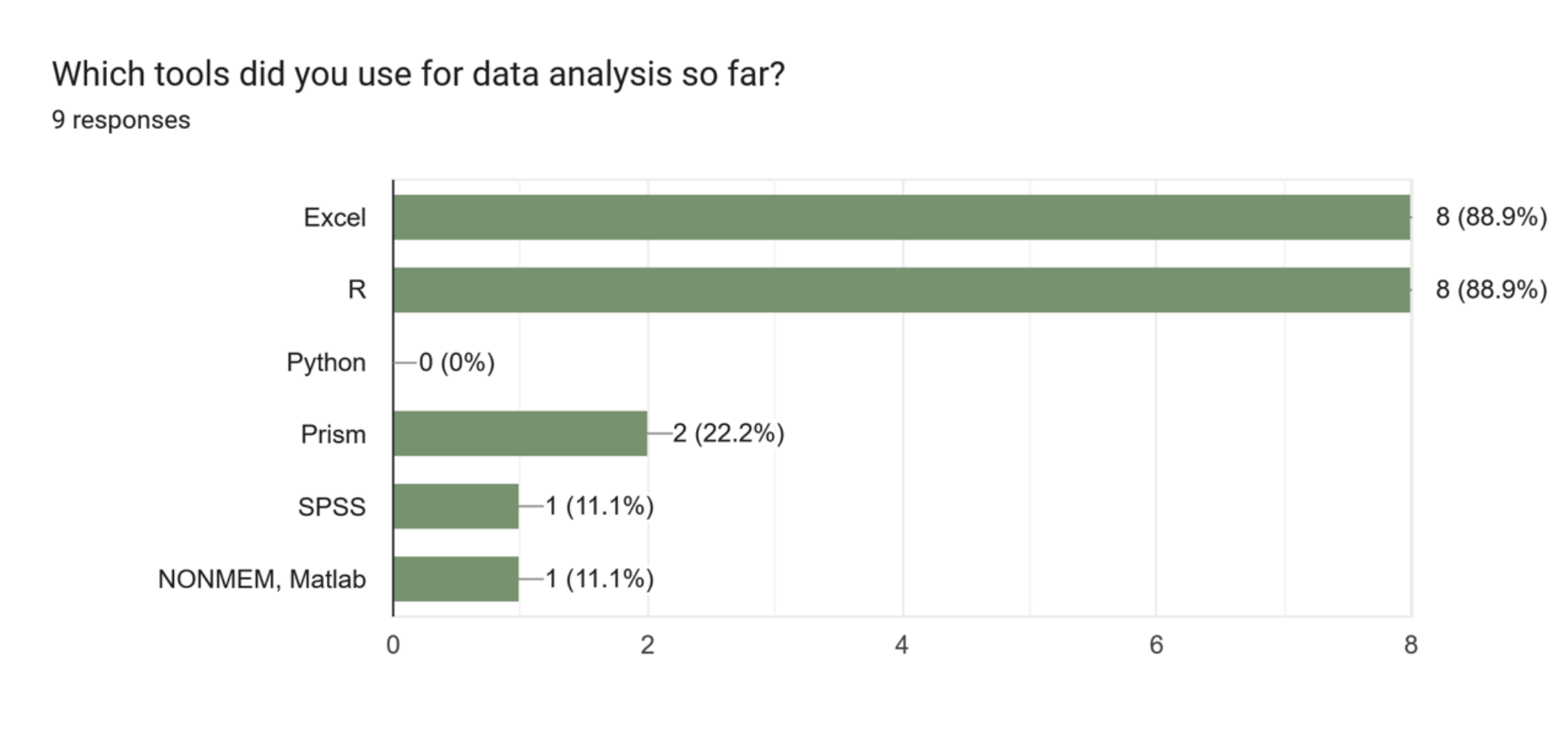
Neurodevelopment

How can model-informed **precision dosing** be applied across different therapeutic areas?

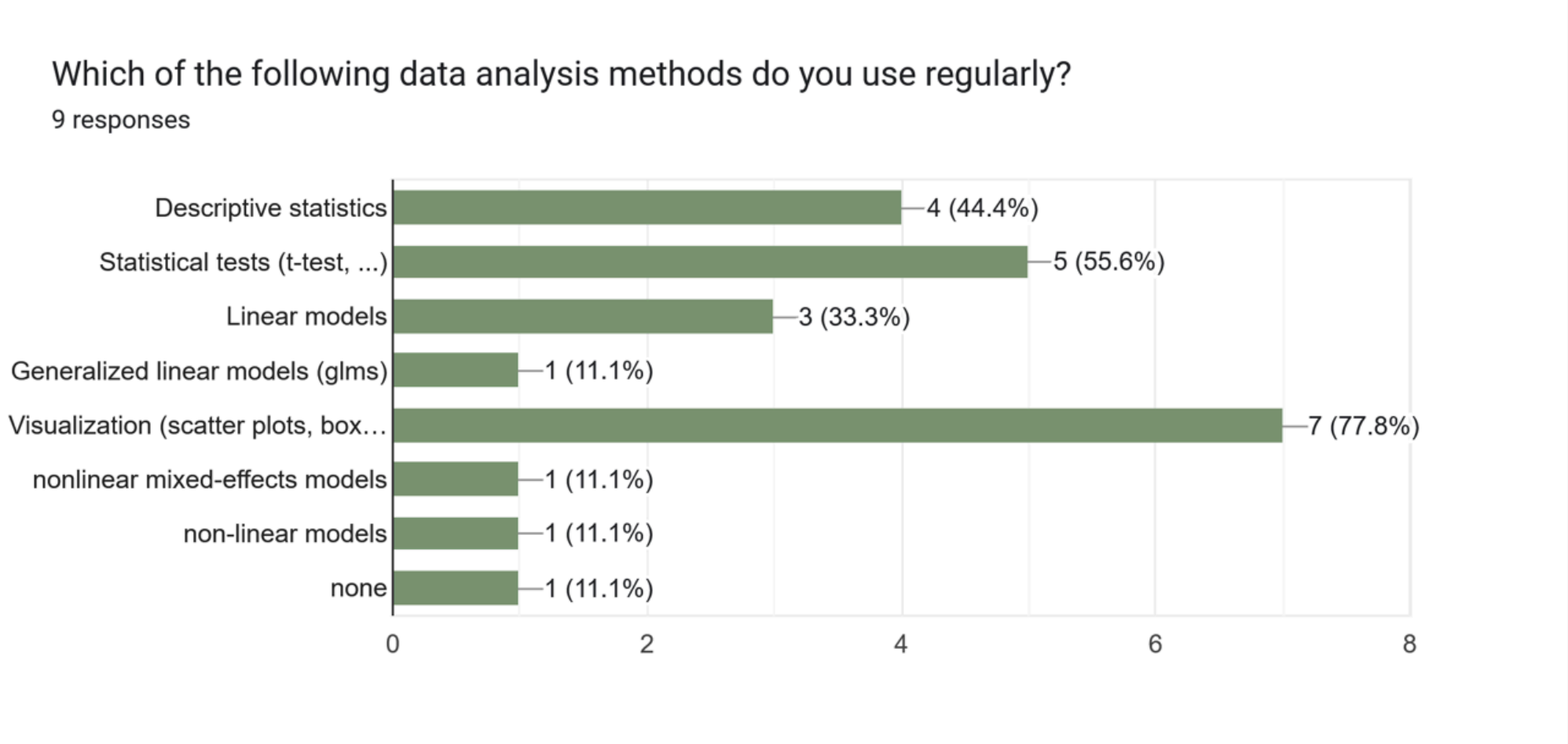
What can **angiosperm inclusions in Baltic amber** tell us about the paleoecosystem and plant evolution?

How can we optimise a specific **antibiotic combination therapy** by using a combined in vitro/in silico approach?

Who are you?


Which tools did you use for data analysis so far?
9 responses
Excel
R
Python
Prism
SPSS
NONMEM, Matlab
8 (88.9%)
8 (88.9%)
0 (0%)
2 (22.2%)
1 (11.1%)
1 (11.1%)
0
2
4
6
8

Who are you?


Which of the following data analysis methods do you use regularly?
9 responses
Descriptive statistics
4 (44.4%)
Statistical tests (t-test, ...)
5 (55.6%)
Linear models
3 (33.3%)
Generalized linear models (glms)
1 (11.1%)
Visualization (scatter plots, box…
7 (77.8%)
nonlinear mixed-effects models
1 (11.1%)
non-linear models
1 (11.1%)
none
1 (11.1%)
0
2
4
6
8

Workshop topics

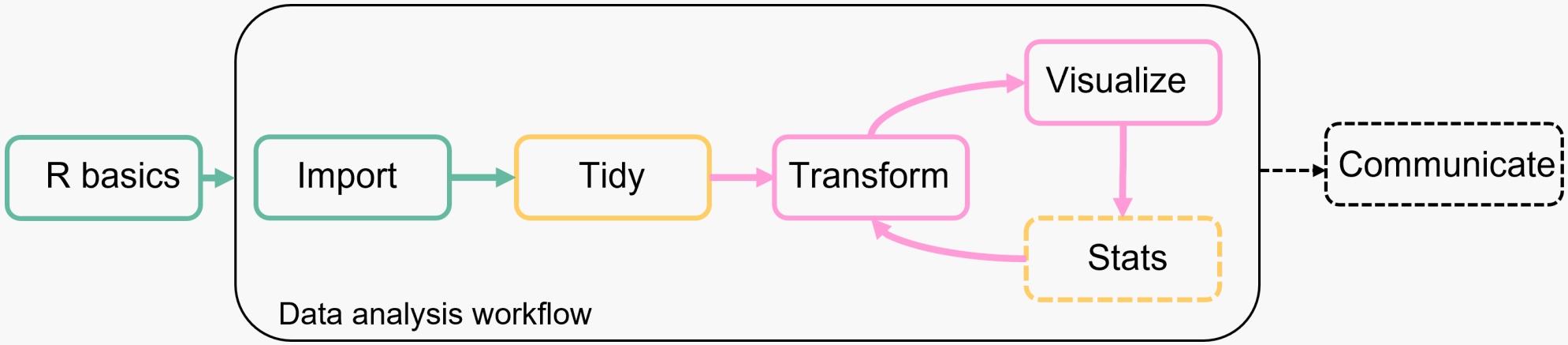


Image adapted from Wickham & Grolemund: R for Data Science

Day1 Introduction to R and RStudio and data import

Day2 Data transformation and visualization with the tidyverse

Day3 Cleaning data, statistical tests, good practice and AI tools

Day 4 Bring your own data

Schedule and Organization

📅 06.10.2025 - 07.10.2025 from 🕗 9 a.m. - 4 p.m.
📅 13.10.2025 - 14.10.2025 from 🕗 9 a.m. - 4 p.m.
🍜 ~ 12 a.m. - 1 p.m., ☕ in between
📌 We will meet in the `General` meeting on Webex

1. Input sessions

- Presentation and demonstration of a topic + examples

2. Tasks

- Solve them in small groups

3. Joint discussion of tasks & questions

Material

- All material can be found on the workshop's website
  - Presentations, Tasks, Solutions, Additional resources
- You can download slides and R Scripts from there
- Website will stay online after the workshop

Bring your own data

On the last workshop day, you can **work with your own research data**. I will also provide some **real life data sets** from different topics.

Learning by doing

- Get started using R for your own analyses
- Use any of the methods from the course or try new things, …
- Present/discuss your results, questions and problems at the end of the day

- Add your name and some details on what you plan to do in this joint table

Before we get started I

- **Help each other** if possible
- Have an eye on the chat and on the time
- All questions and comments are welcome
- If possible, please turn on your camera
- Feedback is welcome (Evaluation at the end of the workshop)

Before we get started II

Did anyone have problems installing R and RStudio?

Download and install R from https://cran.r-project.org

Download and install RStudio from https://www.posit.co

Before we get started III

How to use Webex teams

- `General` channel for our joint meetings and chat
- `Groups 1-4` for solving tasks jointly
  - Group spaces have their own chat and meetings
  - Ask questions in the chat, share screenshots, share your screen and talk

Now you (10 min)

> Get to know your team

- **Go into your groups** and **start a meeting** there
- **Introduce yourself** to your team (~2 minutes for everyone)
  - What is your PhD about?
  - Which tools did you use so far for data analysis?
  - Which types of data analysis do you do/want to do?
- **Come back to the general meeting** after 10 mins